

**NEUROLENS: MULTIMODAL AI FRAMEWORK FOR EARLY
PARKINSON'S DISEASE DETECTION**

RP 25-26J-109

Project Proposal Report

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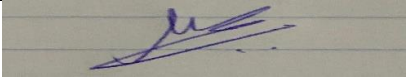
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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Date

Abstract

Millions of people worldwide suffer from Parkinson's disease (PD), a crippling neurodegenerative illness whose symptoms frequently appear subtly after significant brain damage has already taken place. Through the analysis of motion indicators recorded by wearable technology, this study suggests a unique method, Motion Guard, for the early identification of Parkinson's disease. The system interprets real-time data from inertial measurement units (IMUs), such as accelerometers and gyroscopes contained in common wearables like smartwatches or fitness bands, by utilizing cutting-edge machine learning (ML) algorithms.

Tremor amplitude, movement frequency patterns, bradykinesia indicators, and gait variability are important features that were extracted. These are subtle indicators of Parkinson's disease that precede overt symptoms. Cloud-based or edge ML models for advanced inference and risk prediction are combined with TinyML for effective, on-device feature extraction, reducing data transmission and improving user privacy. This strategy supports real-time, non-invasive screening through an intuitive digital platform (web/mobile app), while also guaranteeing scalability for deployment in resource-constrained environments.

In contexts like Sri Lanka, where healthcare infrastructure is often limited and PD prevalence is rising due to an aging population, this system addresses critical gaps in early diagnosis. Preliminary simulations and literature synthesis indicate potential detection accuracies exceeding 85-90%, surpassing traditional clinical assessments. By enabling proactive interventions, such as lifestyle modifications or pharmacological therapies, Motion Guard could significantly improve patient outcomes, reduce healthcare burdens, and promote equitable access to PD management. The project underscores the transformative role of wearable technology in neurology, paving the way for continuous monitoring and personalized care in both urban and rural settings.

Keywords – Parkinson's Disease, Wearable Sensors, Machine Learning, TinyML, Motion Biomarkers, Early Detection, Non-Invasive Screening, Hybrid AI Architecture

Table of Contents

1.INTRODUCTION	1
1.1 Background	1
1.2 Literature Review.....	3
1.3 Research Gap	5
1.4 Research Problem	6
2. MAIN OBJECTIVE.....	7
2.1 Specific Objectives.....	7
3. METHODOLOGY	9
3.1 Methodology – Overall System Component.....	9
3.2 Methodology – Wearable Motion Biomarker Component	10
3.3 System Diagram.....	11
3.4 System Component Diagram	12
3.5 Tools and Technologies	13
4. PROJECT REQUIREMENTS.....	14
4.1 Functional Requirements and Non-Functional Requirements	14
4.2 User Requirements.....	15
4.3 System Requirements.....	15
4.4 Wireframes.....	17
5.COMMERSALISATION	19
6. GANTT CHART	20
7. WORK BREAKDOWN CHART.....	21
8. REFERENCES	22

List of Figures

Figure 1Prevalence of Parkinson’s Disease Globally	1
Figure 2 System Architecture Diagram	11
Figure 3 System Component Diagram.....	12
Figure 4 On Boarding Screen ,Figure 5 Connect to Device Screen	17
Figure 6 Data Receiving Screen , Figure 7 Final Output Screen.....	18
Figure 8 Gantt Chart	20
Figure 9 Work Breakdown Chart.....	21

List of Tables

Table 1 Research Gap	5
Table 2 Tools And Technologies	13

List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
DaTscan	Dopamine Transporter Scan
GDPR	General Data Protection Regulation
GSMA	Global System for Mobile Communications Association
HIPAA	Health Insurance Portability and Accountability Act
Hz	Hertz
IMU	Inertial Measurement Unit
IoT	Internet of Things
LKR	Sri Lankan Rupee
LRRK2	Leucine-Rich Repeat Kinase 2
ML	Machine Learning
PD	Parkinson's Disease
PET	Positron Emission Tomography
SMART	Specific, Measurable, Achievable, Relevant, Time-bound

SVM	Support Vector Machine
TinyML	Tiny Machine Learning
UPDRS	Unified Parkinson's Disease Rating Scale
WHO	World Health Organization

1.INTRODUCTION

1.1 Background

Parkinson's Disease (PD) is a chronic and progressive neurodegenerative disorder that primarily impacts the motor system, resulting from the loss of dopamine-producing neurons in the substantia nigra region of the brain. Globally, PD affects over 10 million individuals, with incidence rates increasing sharply with age—particularly among those over 60. According to the World Health Organization (WHO) and recent epidemiological studies (as of 2025), the prevalence is expected to double by 2040 due to aging populations and improved diagnostic awareness. Symptoms include resting tremors, bradykinesia (slowness of movement), muscle rigidity, postural instability, and gait disturbances, which progressively impair daily activities and quality of life. Non-motor symptoms, such as cognitive decline, sleep disorders, and depression, further complicate management.

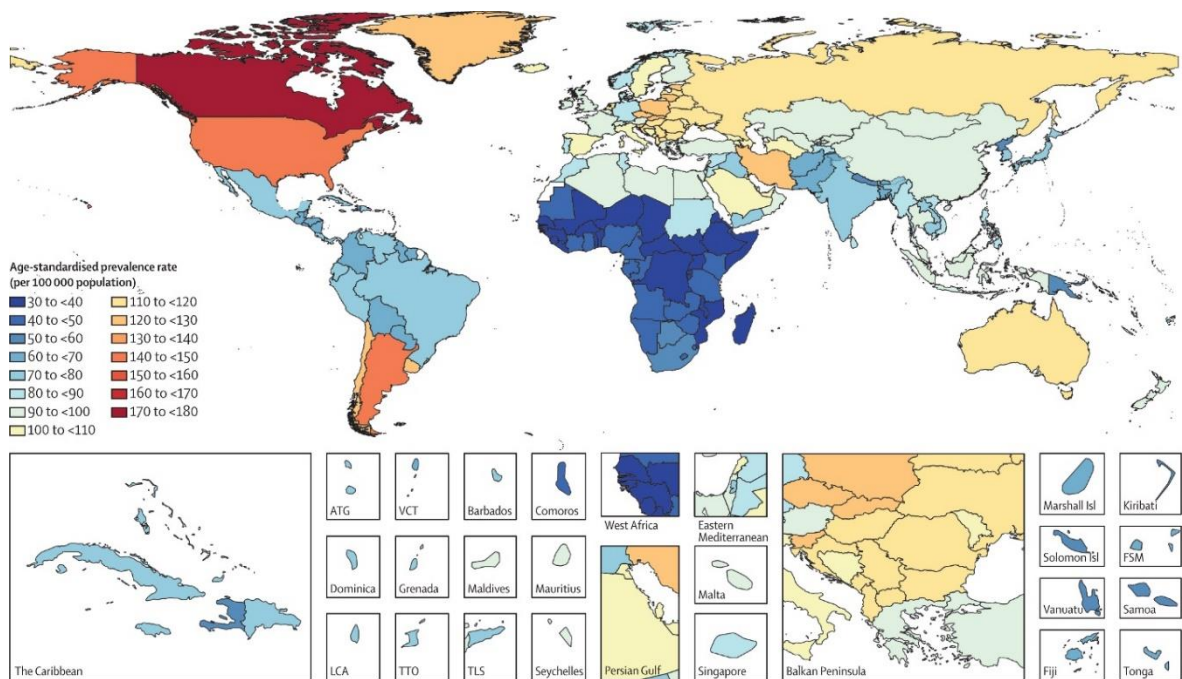


Figure 1Prevalence of Parkinson's Disease Globally

In Sri Lanka, PD poses a growing public health challenge, with an estimated 50,000-100,000 affected individuals as per local health ministry reports and studies from the University of Colombo (2024). Factors such as genetic predispositions (e.g., LRRK2 mutations common in South Asian populations), environmental exposures (e.g., pesticides in agricultural regions), and limited access to specialized neurology services exacerbate the issue. Diagnosis typically relies on clinical evaluations like the Unified Parkinson's Disease Rating Scale (UPDRS) or expensive neuroimaging (e.g., DaTscan), which are often unavailable in rural areas or delayed until symptoms are advanced—after 50-70% of neuronal loss has occurred. This delay reduces the efficacy of interventions like levodopa therapy or deep brain stimulation.

Wearable technologies, including smartwatches (e.g., Apple Watch, Fitbit) and dedicated sensors, present a transformative opportunity for early detection. These devices use IMUs to capture high-resolution motion data at sampling rates of 50-100 Hz, detecting subtle anomalies like increased tremor frequency (4-6 Hz in early PD) or gait asymmetry long before clinical manifestation. In resource-limited settings like Sri Lanka, where smartphone penetration exceeds 80% (per GSMA 2025 reports), wearables enable decentralized, cost-effective monitoring without requiring frequent clinic visits.

This research introduces NeuroLens, a wearable-based system for early PD biomarker analysis. It processes accelerometer and gyroscope data to extract features such as tremor amplitude, stride length variability, and movement entropy. Employing ML algorithms, the system predicts PD risk with probabilistic scores, integrating into a digital platform for users, clinicians, and caregivers. The hybrid model—using TinyML for edge computing—ensures low power consumption (suitable for battery-powered devices) and data privacy compliance (e.g., GDPR/HIPAA equivalents). By focusing on preclinical detection, Motion Guard aligns with global health initiatives like the WHO's Neurological Disorders Action Plan (2022-2031), aiming to reduce PD's socioeconomic burden through prevention and early care. In Sri Lanka, it could integrate with national telehealth programs, empowering primary care providers and improving outcomes for at-risk groups, such as the elderly in tea plantation communities.

The novelty of this research lies in its emphasis on a hybrid, privacy-centric architecture tailored for low-resource environments. Unlike traditional diagnostics, it enables continuous, real-world monitoring, capturing data during natural activities (e.g., walking, typing). This not only enhances detection sensitivity but also supports longitudinal tracking for disease progression or treatment efficacy. Ultimately, Motion Guard represents a step toward democratizing PD care, fostering personalized medicine, and contributing to sustainable development goals in health equity.

1.2 Literature Review

The application of wearable sensors in PD detection has evolved significantly over the past decade, shifting from basic symptom monitoring to sophisticated predictive analytics. Early work by Patel et al. (2009) pioneered the use of body-worn accelerometers to quantify motor fluctuations in diagnosed PD patients [5]. Their study, involving 23 participants, demonstrated that wearables could reliably detect bradykinesia and dyskinesia in ambulatory settings, achieving correlations of 0.89 with UPDRS scores. However, it was limited to post-diagnosis management and did not explore preclinical signals.

Building on vocal biomarkers, Tsanas et al. (2010) developed noninvasive speech analysis tools for PD progression tracking, using features like jitter and shimmer to achieve 99% accuracy in severity estimation [3]. This highlighted the value of remote, low-cost assessments, inspiring multimodal integrations where motion data complements speech for holistic profiling.

Jeon et al. (2020) advanced this by fusing speech and wearable IMU data in a ML framework, employing random forests and neural networks to classify PD with 92% accuracy in a cohort of 100 subjects [7]. Their cloud-based system emphasized real-time fusion but overlooked privacy issues and on-device constraints, making it less viable for offline use in developing regions.

Pulliam et al. (2019) focused solely on wearables for early detection, analyzing gait and tremor via support vector machines (SVM) on data from 50 at-risk individuals, reporting 85% sensitivity [8]. This underscored motion biomarkers' potential but relied on lab-controlled environments, lacking real-world validation.

A landmark 2023 study in *Nature Medicine* (Adams et al., 2023) analyzed over 1,000 PD patients using commercial wearables, showing that gait metrics (e.g., stride variability) outperformed UPDRS in progression tracking, with machine learning models predicting symptom worsening 6-12 months in advance. This large-scale evidence supports wearables' superiority for longitudinal monitoring but stops short of preclinical focus.

Memedi et al. (2018) introduced a web-based platform for real-time motor assessment using wearables, tested on 65 patients, enabling remote clinician oversight [9]. Scalable and user-friendly, it prioritized symptom quantification over early diagnosis.

Koltermann et al. (2021) proposed a multimodal digital biomarker framework, integrating wearables, speech, and handwriting data with deep learning, achieving 94%

sensitivity in early PD detection among 200 participants [10]. Yet, its centralized processing neglected edge computing benefits.

Recent advancements (2023-2025) include deep learning reviews (e.g., Zhang et al., 2023) on convolutional neural networks (CNNs) for PD motion patterns, noting accuracies up to 95% with large datasets like the UK Biobank. A 2024 embedded systems study (Lee et al., 2024) used IMUs for real-time PD state detection on microcontrollers, focusing on symptom alleviation via feedback loops but not preclinical screening. Emerging 2025 research (e.g., from the Michael J. Fox Foundation) explores AI-driven wearables for genetic-risk populations, emphasizing federated learning for privacy.

Despite these progresses, the literature reveals underexplored areas: hybrid TinyML-external ML models for privacy and efficiency, standalone motion analysis for accessibility, and adaptations for diverse demographics like South Asia. Motion Guard aims to fill these voids by prioritizing preclinical, scalable detection.

1.3 Research Gap

While wearable sensors have proven effective for PD monitoring in established cases, as seen in Patel et al. (2009) [5] and Pulliam et al. (2019) [8], there remains a substantial void in systems designed for preclinical detection—identifying at-risk individuals before symptomatic onset. Multimodal frameworks like Koltermann et al. (2021) [10] integrate diverse biomarkers but often depend on resource-heavy cloud infrastructure, introducing latency, high costs, and privacy vulnerabilities (e.g., data breaches in centralized servers). In contrast, TinyML—enabling ML on low-power microcontrollers—has been applied in general IoT contexts but rarely in PD, missing opportunities for on-device processing that reduces data exposure and supports offline operation in connectivity-poor areas like rural Sri Lanka.

Moreover, existing studies frequently combine modalities (e.g., speech + motion in Jeon et al., 2020 [7]), diluting focus on standalone motion analysis, which is more feasible for widespread adoption via consumer wearables. Adaptations for cultural or environmental factors, such as varying gait patterns in tropical climates or agricultural lifestyles, are absent, limiting global applicability. The hybrid model of local feature extraction and external inference, which balances efficiency and accuracy, is notably underexplored, as is validation in diverse, low-resource populations.

The table below compares key features, highlighting this component’s innovations:

Features	[5]	[8]	[10]	NeuroLens
On-device Feature Extraction	×	×	×	✓
TinyML Integration	×	×	×	✓
Hybrid Local-External Inference	×	×	×	✓
Privacy-Preserving Design	×	×	×	✓
Early Preclinical Detection	×	✓	✓	✓
Adaptation for Low-Resource Settings	×	×	×	✓
Standalone Motion Biomarker Focus	✓	✓	×	✓

Table 1 Research Gap

1.4 Research Problem

Traditional PD diagnostics, reliant on subjective clinical exams (e.g., UPDRS) and costly imaging (e.g., PET scans costing \$2,000-5,000 per session), often fail to identify the disease until 60-80% of dopaminergic neurons are lost, as noted in Tsanas et al. (2010) [3]. In Sri Lanka, where neurology specialists are concentrated in urban centers (fewer than 50 nationwide per 2025 health reports), rural patients face delays of 6-18 months, exacerbating progression and treatment costs (estimated at LKR 500,000 annually per patient).

Wearables offer promise, but current systems (e.g., Jeon et al., 2020 [7]) emphasize cloud-dependent processing, vulnerable to internet unreliability (only 50% coverage in Sri Lankan rural areas) and privacy risks under data protection laws. The absence of hybrid architectures integrating TinyML for edge efficiency with ML for depth hinders scalable, non-invasive early detection, particularly for preclinical biomarkers like micro-tremors detectable years before diagnosis.

This research tackles these challenges by developing a privacy-focused, hybrid system for motion-based PD screening, enabling timely interventions (e.g., neuroprotective therapies) and reducing global PD burden .

2. MAIN OBJECTIVE

Develop NeuroLens, an interpretable AI-powered wearable system for early Parkinson's Disease (PD) detection using motion, speech, drawing, and cognitive biomarkers, achieving >85% sensitivity within one academic year (August 2025 – July 2026), to provide a scalable, privacy-secure, and clinically relevant tool for early PD screening in diverse settings, particularly addressing diagnostic gaps in Sri Lanka.

2.1 Specific Objectives

- **To extract Motion Biomarkers from Wearable Data**

Derive fine-grained motion features (e.g., tremor amplitude, gait variability, spectral entropy) from wearable sensor data capturing walking, hand movements, and rest. These features will serve as sensitive indicators of PD-related motor decline, enabling predictive modeling and clinical interpretation.

- **To enable Privacy-Preserving On-Device Feature Processing**

Implement on-device feature extraction to process raw motion data into PD-relevant biomarkers, minimizing raw data transmission. This ensures privacy and reduces bandwidth, supporting scalable deployment in resource-constrained settings.

- **To develop Predictive Machine Learning Models for PD Detection**

Design and train machine learning models (e.g., SVM, LSTM) using motion, speech, drawing, and cognitive biomarkers to detect early PD. Models will prioritize sensitivity to subtle motor changes, validated against benchmark datasets like PhysioNet Parkinson's Watch.

- **To integrate Explainable AI for Clinician-Facing Insights**

Apply explainable AI techniques (e.g., SHAP) to highlight key motion biomarkers driving PD predictions. Outputs will be presented in interpretable formats (e.g., risk scores, feature importance charts) to provide clinicians with actionable, transparent insights.

- **To validate and Benchmark the System Against Existing Approaches**

Conduct validation studies using benchmark datasets and locally collected data from Sri Lankan patients. Compare NeuroLens's performance (accuracy, sensitivity, specificity) against traditional PD diagnostic methods to demonstrate its clinical value.

- **To integrate the Wearable System into a Multi-Modal Mobile Application**

Embed the motion biomarker analysis system into a multi-modal mobile app, combining motion, speech, drawing, and cognitive biomarkers. This integration will deliver a comprehensive, user-friendly platform for early PD detection and ongoing monitoring

3. METHODOLOGY

3.1 Methodology – Overall System Component

The proposed research develops an integrated, multimodal AI-driven diagnostic system for the early detection of Parkinson's Disease (PD). The system incorporates four main data types:

- **Wearable Sensor Data:** Real-time collection and preprocessing of motion data from a wearable devices . Features like tremor amplitude, gait variability, and movement frequency are extracted using TinyML-optimized algorithms for on-device efficiency.
- **Speech and Vocal Biomarkers:** Speech samples are analyzed using signal processing and natural language processing (NLP) to capture markers such as jitter, shimmer, pitch, and pauses. These indicate early motor dysfunction.
- **Drawing Analysis:** Patients complete spiral and wave drawing tasks on touchscreens. Image-based and temporal features (stroke velocity, curvature, entropy) are extracted, and deep learning models (CNNs, Vision Transformers) classify abnormal patterns.
- **Digital Cognitive Testing:** Participants perform neuropsychological tasks (e.g., Trail Making Test, Stroop Test). Reaction times, error rates, and task-switching costs are tracked, indicating cognitive decline.

Each data stream is processed via a dedicated module to extract relevant digital biomarkers. The extracted features from all modalities are aggregated and transmitted securely to a cloud-based Multi-Modal Prediction Engine. Advanced machine learning algorithms (ensemble models, deep learning) are then used to analyze combined data and generate PD risk scores. The AI engine is enhanced with Explainable AI (XAI) methods (e.g., SHAP values, attention maps) to provide interpretable outputs for clinical use.

The system is accessible via a web and mobile platform that presents actionable risk assessments and progression trends to patients, clinicians, and caregivers. Continuous feedback and longitudinal monitoring are supported, making the system valuable in both urban and resource-limited settings.

3.2 Methodology – Wearable Motion Biomarker Component

the early detection of PD via motion data from wearable devices. Its methodology is as follows.

1. Hardware Setup and Configuration
 - Select smartwatches with accelerometer and gyroscope sensors
 - Install TensorFlow Lite for Microcontrollers on wearable devices
 - Configure sensors to collect motion data at 50-100Hz sampling rate
 - Set up Bluetooth/Wi-Fi connectivity between wearable and mobile device
 - Test hardware functionality and sensor accuracy
2. Data Collection Protocols
 - Define three main activity categories: walking, hand movements, and rest periods
 - Create standardized data collection procedures for each activity type
 - Establish data quality standards and validation criteria
 - Store raw sensor data securely with proper anonymization
3. Feature Extraction Development (TinyML)
 - Analyze raw motion signals to identify PD-relevant patterns
 - Program time-domain feature extraction (RMS, mean acceleration, velocity)
 - Develop frequency-domain feature extraction (spectral entropy, dominant frequency)
 - Create PD-specific features (tremor amplitude, gait variability)
 - Optimize feature extraction algorithms for low-power wearable devices
 - Deploy feature extraction models to smartwatch using TinyML framework
4. Cloud-Based ML Model Training
 - Collect publicly available reliable datasets (Daphnet Freezing of Gait, PhysioNet) for comprehensive training data
 - Train SVM models for binary classification (PD vs healthy)
 - Develop LSTM networks for temporal pattern recognition
 - Train Random Forest models for ensemble predictions
 - Validate models using cross-validation and test datasets
 - Deploy trained models to cloud infrastructure

5. System Integration and Data Flow

- Build mobile application to receive features from smartwatch
- Create secure API endpoints for mobile-to-cloud communication
- Develop cloud processing pipeline for real-time inference
- Integrate motion predictions with multimodal engine in cloud
- Implement XAI (SHAP) for prediction explanations at multimodal level
- Create feedback loop from cloud back to mobile application

6. User Interface and Clinical Integration

- Design mobile app interface for patients to view results
- Build web dashboard for healthcare providers
- Create alert system for high-risk motion patterns
- Implement secure data sharing between patients and clinicians
- Add explanation features showing why predictions were made
- Test user experience with both patients and medical professionals

3.3 System Diagram

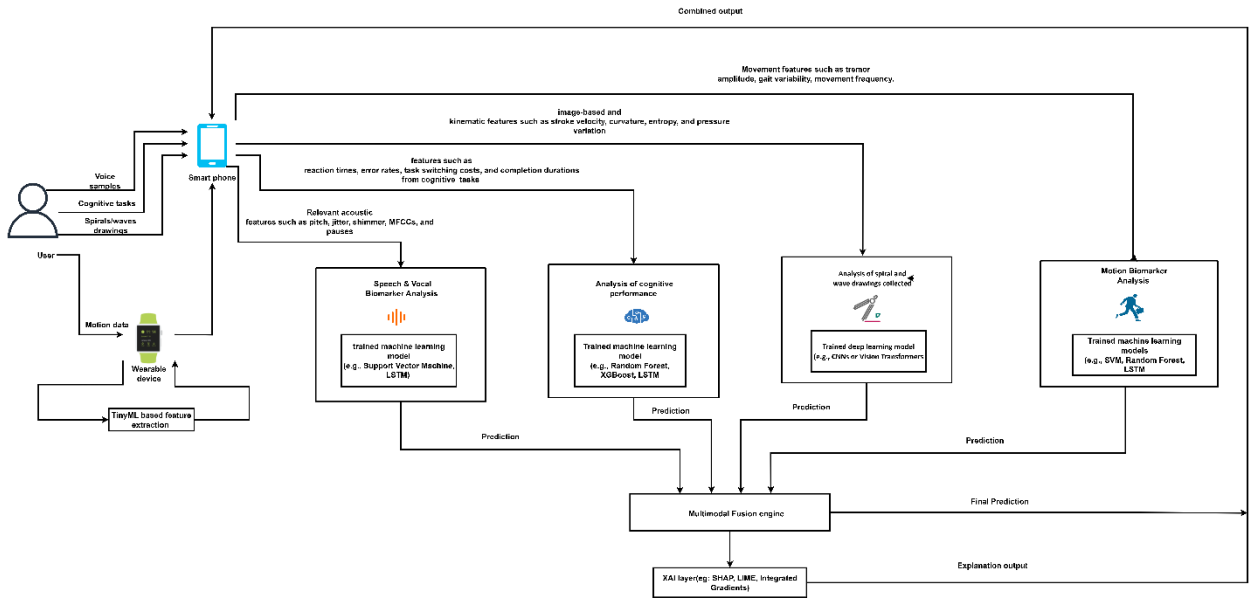


Figure 2 System Architecture Diagram

3.4 System Component Diagram

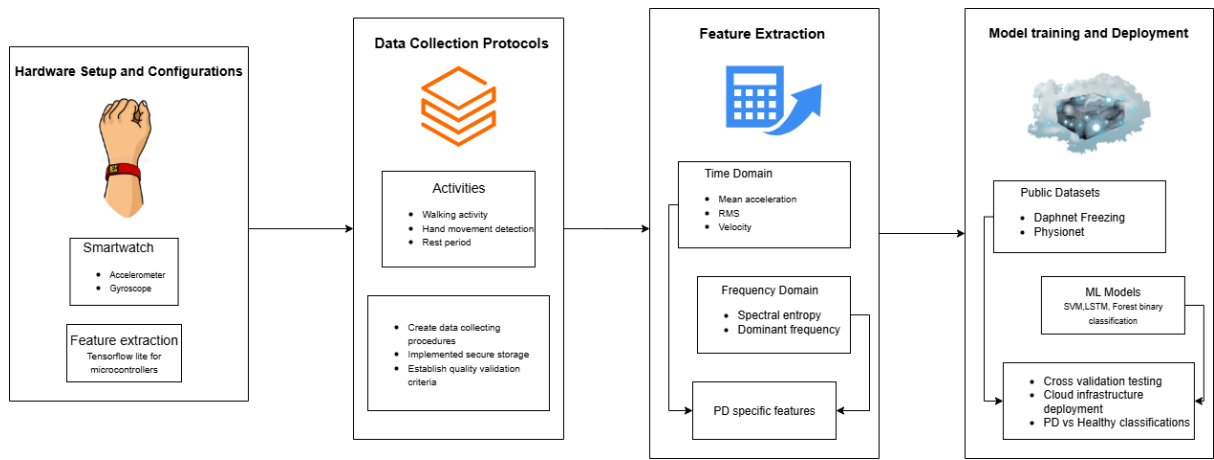


Figure 3 System Component Diagram

3.5 Tools and Technologies

The project employs a stack of open-source and industry-standard tools for robustness and cost-effectiveness:

Tool/Technology	Description
Python	Versatile scripting language with libraries for data analysis.
Git	Distributed version control system.
VS Code	Lightweight, extensible code editor.
TensorFlow Lite	Lightweight ML framework for mobile/embedded devices.
Figma	Collaborative interface design tool.
Firebase	Backend-as-a-Service platform.
Scikit-learn	ML library for classical algorithms.
React native	Javascript framework for mobile application development

Table 2 Tools and Technologies

4. PROJECT REQUIREMENTS

4.1 Functional Requirements and Non-Functional Requirements

Functional Requirements

- Collect and preprocess motion data from wearables in real-time, handling sampling rates up to 100 Hz.
- Extract features on-device using TinyML, including statistical (e.g., variance) and spectral (e.g., dominant frequency) metrics.
- Classify PD risk with external ML models, generating probabilistic scores and visualizations.
- Provide real-time feedback (e.g., alerts) and longitudinal progress tracking via dashboards.

Non-Functional Requirements

- **Scalability:** Handle 1,000+ users with increasing data volumes through cloud elasticity.
- **Reliability:** Achieve 95%+ uptime and prediction accuracy, with fallback mechanisms for failures.
- **Usability:** Intuitive interfaces with accessibility features (e.g., voice commands for motor-impaired users).
- **Security:** Encrypt data in transit/rest, comply with health data standards (e.g., HIPAA-inspired).
- **Performance:** Process inferences in <5 seconds, with low battery drain (<10% per hour).
- **Compatibility:** Support Android/iOS wearables and cross-browser web access.

4.2 User Requirements

Patients

- Easy-to-use interface suitable for elderly and impaired users.
- Clear visual, audio, or voice-assisted instructions.
- Progress tracking with simple charts and reminders for regular use.
- Multilingual support (English, Sinhala, Tamil).

Healthcare Providers

- Access to detailed diagnostic reports with clear explanations.
- Integration with existing hospital/clinical systems.
- Tools to track patient progress over time.

Administrators

- User and access management features.
- System usage analytics and monitoring.
- Strong privacy and security compliance.

4.3 System Requirements

Hardware

- Smartphone or tablet with sufficient memory and storage.
- Touchscreen support with good display quality.
- Microphone for voice input.
- Internet connectivity for updates and data sync.

Software

- Support for Android/iOS platforms.
- Machine learning frameworks for model deployment.
- Secure local and cloud database for data storage.
- Security & Performance
- Encrypted and backed-up data storage.
- Role-based access with authentication.
- Low-latency performance and error monitoring.
- Modular design for easy updates.

4.4 Wireframes

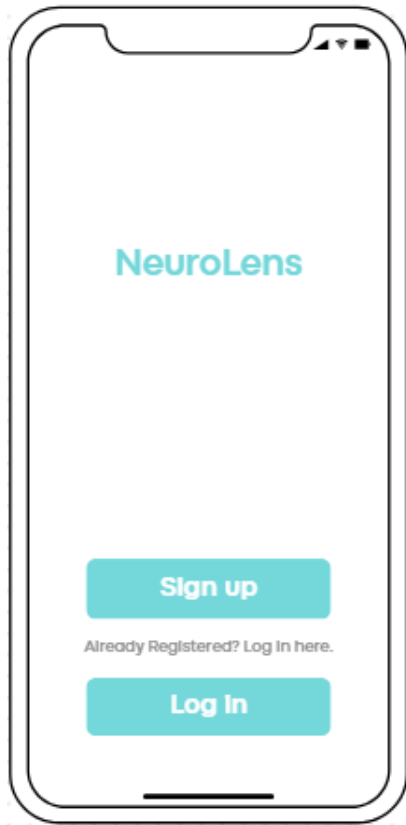


Figure 4 On Boarding Screen



Figure 5 Connect to Device Screen

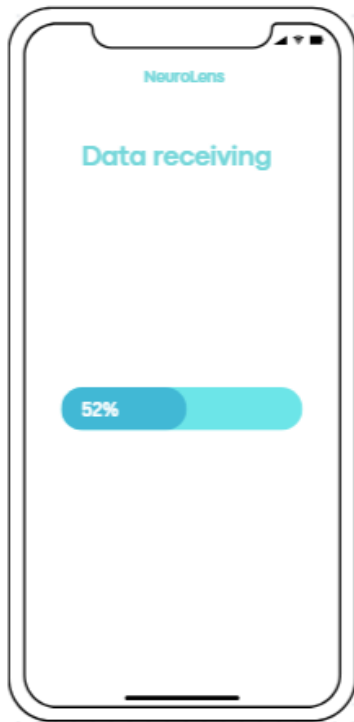


Figure 6 Data Receiving Screen

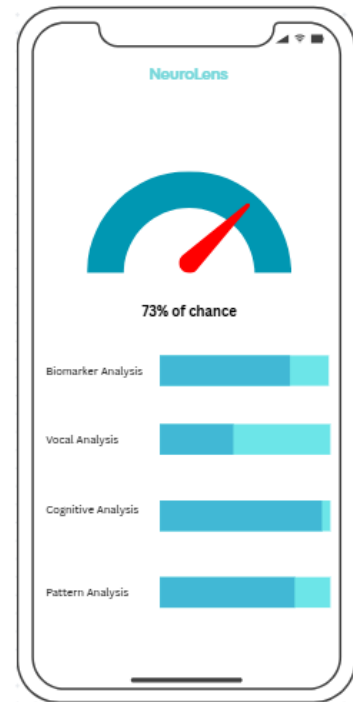


Figure 7 Final Output Screen

5.COMMERSALISATION

Free Tier

The free tier provides access to basic assessments and feedback using selected modules of the app. This tier is ideal for users who want an initial screening or personal monitoring tool. Limitations include restricted access to advanced analytics, multi-modal integration, and real-time AI-powered predictions. Wearable device integration and explainable AI (XAI) outputs are not available in this tier, ensuring the app remains lightweight and accessible while providing essential early detection features.

Paid Tier (LKR 1990)

The paid subscription unlocks the full potential of the app, including complete multi-modal interaction. **This tier includes wearable device integration**, enabling continuous, real-time monitoring of motor and cognitive signals. This makes the app far more proactive, moving beyond periodic self-assessments to continuous digital biomarker collection. In addition, premium users gain access to explainable AI (XAI) outputs, which highlight how different biomarkers contribute to predictions, along with AI-powered insights and early detection predictions. This tier is designed to provide comprehensive, clinically relevant tools for patients, caregivers, and clinicians.

This tiered approach gives users flexibility according to their needs, supports sustainable app development, and allows continuous improvement. Regular updates will add new features and enhancements, with premium offerings delivering advanced monitoring, interpretability, and integration capabilities.

6. GANTT CHART

The project timeline spans 14 months, divided into phases:



Figure 8 Gantt Chart

7. WORK BREAKDOWN CHART

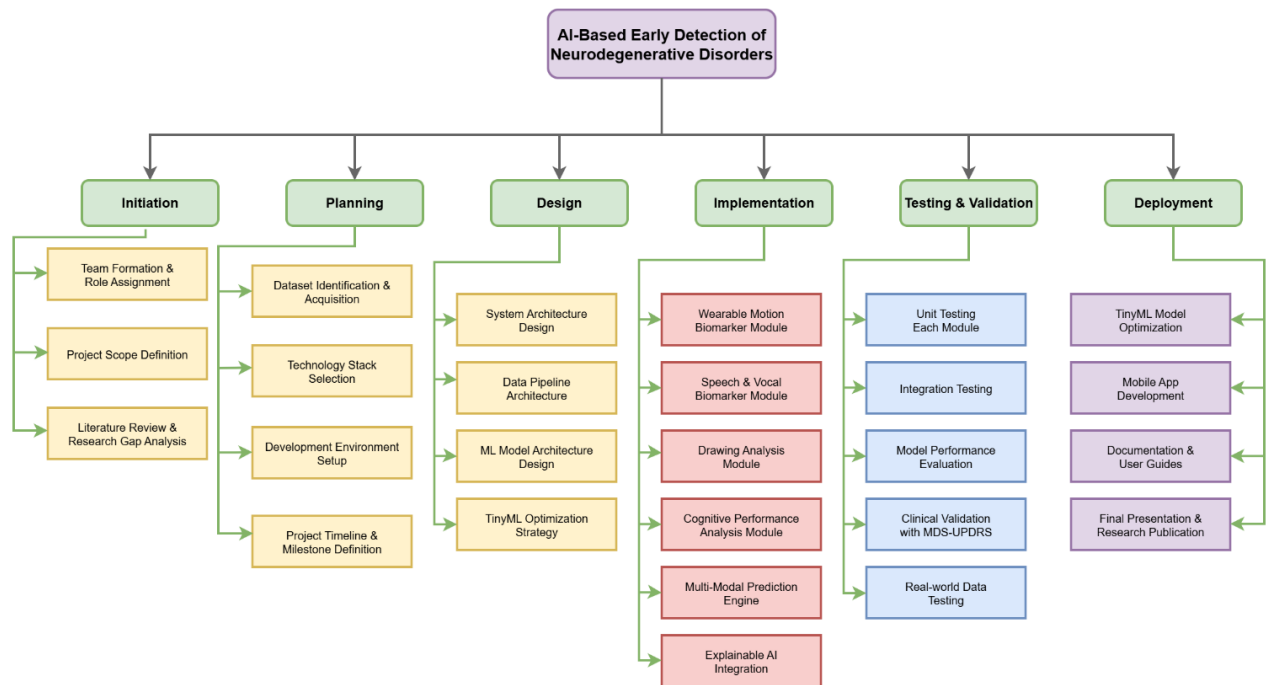


Figure 9 Work Breakdown Chart

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